ATMA RAM SANATAN DHARMA COLLEGE

ARTIFICIAL INTELLIGENCE: PRACTICAL FILE

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Course: BSC (Hons.) Computer Science

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1. Family Tree

```
male(ramesh).   male(suresh).   male(mahesh).
male(rajesh).   male(vikram).   male(rohan).

female(sunita).  female(kavita).
female(priya).   female(anita).  female(pooja).

parent(ramesh, suresh).   parent(ramesh, mahesh).
parent(sunita, suresh).   parent(sunita, mahesh).
parent(suresh, rajesh).   parent(suresh, priya).
parent(kavita, rajesh).   parent(kavita, priya).
parent(mahesh, vikram).   parent(mahesh, anita).

father(X, Y)      :- parent(X, Y), male(X).
mother(X, Y)      :- parent(X, Y), female(X).
grandparent(X, Z) :- parent(X, Y), parent(Y, Z).
sibling(X, Y)     :- parent(Z, X), parent(Z, Y), X \= Y.
brother(X, Y)     :- sibling(X, Y), male(X).
sister(X, Y)      :- sibling(X, Y), female(X).
uncle(X, Y)       :- parent(Z, Y), brother(X, Z).
aunt(X, Y)        :- parent(Z, Y), sister(X, Z).
cousin(X, Y)      :- parent(A, X), parent(B, Y), sibling(A, B).
ancestor(X, Y)    :- parent(X, Y).
ancestor(X, Y)    :- parent(X, Z), ancestor(Z, Y).
```

Outputs:

father(ramesh, suresh).

true

mother(sunita, mahesh).

true

cousin(rajesh, vikram).

true

parent(ramesh, X).

X = suresh
X = mahesh

grandparent(ramesh, X).

X = rajesh
X = priya
X = vikram
X = anita

2. Concatenate Lists

```
conc([], L, L).
conc([H|T], L2, [H|L3]) :- conc(T, L2, L3).
```

Output:

conc([1,2,3], [4,5], L3).

L3 = [1, 2, 3, 4, 5]

3. Reverse List

```
reverse([], []).
reverse([H|T], R) :- reverse(T, RT), conc(RT, [H], R).

conc([], L, L).
conc([H|T], L2, [H|L3]) :- conc(T, L2, L3)
```

Output:

reverse([1,2,3,4], R).

R = [4, 3, 2, 1]

4. Sum of Two Numbers

```
sum(X, Y, S) :- S is X + Y.
```

Output:

sum(10, 25, S).

S = 35

5. Find MAX Number

```
max(X, Y, X) :- X >= Y.
max(X, Y, Y) :- Y > X.
```

Output:

max(3, 7, M).

M = 7

6. Factorial

```
factorial(0, 1).
factorial(N, F) :-
    N > 0,
    N1 is N - 1,
    factorial(N1, F1),
    F is N * F1.
```

Output:

factorial(0, F).

F = 1

7. Fibonacci Series

```
generate_fib(0, 0).
generate_fib(1, 1).
generate_fib(N, T) :-
    N > 1,
    N1 is N - 1,
    N2 is N - 2,
    generate_fib(N1, T1),
    generate_fib(N2, T2),
    T is T1 + T2.
```

Output:

generate_fib(10, T).

T = 55

8. Power of a Number

```
power(_, 0, 1).
power(Num, Pow, Ans) :-
    Pow > 0,
    Pow1 is Pow - 1,
    power(Num, Pow1, Ans1),
    Ans is Num * Ans1.
```

Output:

power(2, 10, Ans).

Ans = 1024

9. Multiplication

```
multi(N1, N2, R) :- R is N1 * N2.
```

Output:

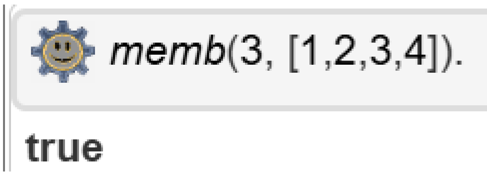
multi(6, 7, R).

R = 42

10. Member of List?

```
memb(X, [X|_]).
memb(X, [_|T]) :- memb(X, T).
```

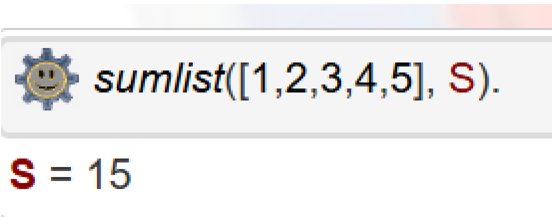
Output:



11. Sum of List

```
sumlist([], 0).
sumlist([H|T], S) :-
    sumlist(T, S1),
    S is H + S1.
```

Output:



12. Even length or Odd length list?

```
evenlength([]).
evenlength([_,_|T]) :- evenlength(T).

oddlength([_]).
oddlength([_,_|T]) :- oddlength(T).
```

Output:

```
evenlength([1,2,3,4]).
true
```

```
oddlength([1,2,3,4,5]).
true
```

13. Max of List

```
maxlist([X], X).
maxlist([H|T], H) :- maxlist(T, M), H >= M.
maxlist([H|T], M) :- maxlist(T, M), M > H.
```

Output:

```
maxlist([3,1,4,1,5,9,2,6], M).
M = 9
```

14. Insert into List

```
insert(I, 1, L, [I|L]).
insert(I, N, [H|T], [H|R]) :-
    N > 1,
    N1 is N - 1,
    insert(I, N1, T, R).
```

Output:

```
insert(99, 1, [1,2,3], R).
R = [99, 1, 2, 3]
```

15. Delete from List

```
delete(1, [_|T], T).
delete(N, [H|T], [H|R]) :-
    N > 1,
    N1 is N - 1,
    delete(N1, T, R).
```

Output:

```
delete(1, [a,b,c,d], R).
R = [b, c, d]
```